DRAWING DESCRIPTIONS — PATENT B

Configurable Recursive Self-Observation Method and System for Spiking Neural Networks with Dynamic Reflection Coefficient Modulation

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block diagram of the self-observation feedback loop showing how the aggregate output of the spiking neural network at timestep t is fed back as additional input at timestep t+1.

FIG. 2 is a spectrum diagram showing the behavioral effects of the reflection coefficient across its full range from 0.0 to 1.0.

FIG. 3 is a block diagram showing the dual modulation sources that dynamically control the reflection coefficient value.

FIG. 4 is a flow diagram of the self-referential learning loop combining STDP synaptic plasticity with recursive self-observation.

FIG. 5 is a control diagram showing energy-aware self-observation regulation where the reflection coefficient is modulated based on system energy state.

FIG. 6 is an end-to-end signal flow diagram showing how the self-observation mechanism integrates within the complete self-sustaining cognitive loop, distinguishing the self-observation feedback pathway from the energy feedback pathway and the synaptic computation pathway.

DETAILED DESCRIPTION OF THE DRAWINGS

Reference Numeral Convention: Each figure in this patent uses a per-figure numbering series (FIG. 1 uses 100-series, FIG. 2 uses 200-series, through FIG. 6 uses 600-series), with even-numbered increments. Where the same physical component appears in multiple figures, it receives the numeral appropriate to that figure's series.

FIG. 1 — Self-Observation Feedback Loop

FIG. 1 is a block diagram illustrating the core self-observation feedback loop of the present invention. An external input 100 carrying signal I_ext(t) enters the diagram from the left and is received by a summing junction 102.

The summing junction 102 combines the external input 100 with a reflection signal to produce an adjusted input I_adj = I_ext + I_refl, which is then delivered to a spiking neural network 104 comprising N leaky integrate-and-fire (LIF) neurons. Within the spiking neural network 104, each neuron updates its membrane potential according to V_i(t+1) = V_i(t) x leak + I_adjusted x dt.

The spiking neural network 104 produces a spike output 106 carrying signal S(t), which exits the diagram to the right and represents the discrete firing events of the network. Simultaneously, the spiking neural network 104 produces an aggregate output 108 defined as O(t) = mean(V_1, V_2, ..., V_N), which is the mean membrane potential across all N neurons. The aggregate output 108 captures the collective internal state of the network at each timestep.

The aggregate output 108 feeds into a self-observation feedback path 114, shown

as a bold line returning along the bottom of the diagram. Along this self-observation feedback path 114, the signal first passes through a delay element 110, which imposes a one-timestep delay (z^-1) ensuring that the network observes its state from timestep t when processing timestep t+1. The delayed signal then enters a reflection scaling block 112, which computes the reflection input as I_refl = O(t) x reflection_coeff. The output of the reflection scaling block 112 is routed upward to the summing junction 102, closing the self-observation loop.

The self-observation feedback path 114 is the central innovation depicted in FIG. 1: the network observes its own prior aggregate state and incorporates that observation as additional input on the next processing step.

FIG. 2 — Reflection Coefficient Spectrum

FIG. 2 is a spectrum diagram showing the behavioral effects of the reflection coefficient across its full operating range. A main spectrum bar 200 spans the

full width of the figure, with the left end labeled "0.0" and the right end labeled "1.0." The current operational range of 0.1 to 0.3 is marked on the main spectrum bar 200.

The main spectrum bar 200 is divided into five distinct behavioral regions.

A first region 202 occupies the leftmost position at 0.0 on the main spectrum bar 200. The first region 202 is labeled "No self-observation" and represents pure feedforward processing in which the system responds only to external input with no self-referential component.

A second region 204 spans from 0.0 to 0.2 on the main spectrum bar 200. The second region 204 is labeled "Minimal self-awareness" and represents a mode in which the system exhibits slight influence of prior state but remains primarily externally driven.

A third region 206 spans from 0.2 to 0.4 on the main spectrum bar 200 and is identified as the "OPERATIONAL RANGE." The third region 206 is labeled

"Balanced self-observation" and represents the preferred operating regime in which external input and meaningful self-reference are combined. The default reflection coefficient of 0.2 + modulation x 0.1 falls within the third region 206.

A fourth region 208 spans from 0.4 to 0.7 on the main spectrum bar 200. The fourth region 208 is labeled "Self-dominated processing" and represents a mode in which internal state dominates external input, creating risk of echo chamber dynamics.

A fifth region 210 spans from 0.7 to 1.0 on the main spectrum bar 200. The fifth region 210 is labeled "Self-absorption" and represents a mode in which external input is overwhelmed by the system's own prior state, causing the system to lock onto its internal dynamics.

Below the main spectrum bar 200, an arrow indicates the direction of increasing self-referential processing from left to right. A note identifies the

reflection coefficient as the system's self-awareness dial, continuously adjustable across the full range depicted by the main spectrum bar 200.

FIG. 3 — Dynamic Modulation Sources

This figure shows two modulation pathways converging on the reflection coefficient computation in a vertical stacked layout.

Reference numerals: 300, 302, 304, 306, 308, 310, 312

Top element (302):

- Cloud shape labeled "EXTERNAL COGNITIVE CONTROL LAYER"

- Inside: "Claude or other cognitive agent provides modulation"

- Arrow down to computation block labeled "modulation (-1.0 to +1.0)"

Center element (300):

- Rectangle with bold border labeled "REFLECTION COEFFICIENT COMPUTATION"

- Inside: "reflection_coeff = base_coeff + modulation x sensitivity"

- Default values: "base_coeff = 0.2, sensitivity = 0.1"

Output (310):

- Arrow from computation block going right

- Reference numeral 310 above arrow

Below computation (side by side):

- Left box (312): "SOURCE PRIORITY" - "External overrides Internal when active"

- Right box (306): "ENERGY STATE MONITOR" - "Reads energy_mwh from EnergyHarvester"

and "Applies energy-aware modulation rules"

- Arrow up from Energy Monitor to Computation labeled "energy-aware modulation"

Bottom callout boxes (side by side):

- Left box (304): "External Modulation Examples"

- "-0.5: Reduce self-observation (focus outward)"

- "0.0: Neutral (use base coefficient)"

- "+0.8: Increase self-observation"

- Right box (308): "Energy Rules"

- "<15 mWh: -0.4"

- "<30 mWh: -0.1"

- ">80 mWh: +0.3"

- "else: 0.0"

FIG. 4 — Self-Referential Learning Loop (STDP + Self-Observation)

This figure shows how STDP learning interacts with the self-observation signal.

Reference numerals: 400, 402, 404, 406, 408, 410

Vertical flow (top to bottom):

Top (400):

- Full-width rectangle labeled "SNN PROCESSING STEP t" (bold border)

- Inside: "Membrane dynamics + spikes"

- Arrow down labeled "S(t)"

Second row (402):

- Full-width rectangle labeled "STDP WEIGHT UPDATE"

- Inside:

- "Post fires: W += 0.01 × pre_trace"

- "Pre fires: W -= 0.012 × post_trace"

- "A-/A+ = 1.2 (stability ratio)"

- Arrow down

Third row (404):

- Full-width rectangle labeled "UPDATED WEIGHT MATRIX"

- Inside: "W(t+1) bounded to [0.0, 0.5]"

- Arrow down to Self-Observation

Fourth row (side by side):

- Left (406): Rectangle labeled "SELF-OBSERVATION"

- Inside: "O(t) × refl_coeff"

- Right (408): Rectangle labeled "KEY INSIGHT"

- Inside: "STDP strengthens causal connections. Self-observation adds correlated feedback."

Feedback loop:

- Bold line from Self-Observation (406) left side, up, and back to SNN Processing (400)

- Vertical text: "FEEDBACK LOOP"

Fifth row (410):

- Full-width rectangle labeled "VALIDATION RESULTS (Phase 8)"

- Inside:

- "F8.1: Entropy 2.95 to 2.02 bits (PASS)"

- "F8.2: STDP MI = 6.52x > 1.20x (PASS)"

Bottom:

- Signal Pathways legend box

- Bold line sample with label: "Bold: Self-observation feedback loop"

FIG. 5 — Energy-Aware Self-Observation Regulation

This figure shows a control diagram for energy-based reflection modulation.

Reference numerals: 500, 502, 504, 506

Top — Energy State Bar (500):

- Full-width horizontal bar labeled "SYSTEM ENERGY LEVEL (mWh)"

- Four labeled regions with dividers:

- 0-15 mWh: CRITICAL

- 15-30 mWh: LOW

- 30-80 mWh: NORMAL

- 80-100 mWh: SURPLUS

- Scale markers below: 0, 15, 30, 80, 100

Second row — Modulation Mapping (502):

- Full-width box labeled "MODULATION MAPPING"

- State labels above each arrow: CRITICAL, LOW, NORMAL, SURPLUS

- Four arrows pointing down to modulation values:

- CRITICAL -> -0.4

- LOW -> -0.1

- NORMAL -> 0.0

- SURPLUS -> +0.3

Third row — Reflection Coefficient Scale (504):

- Full-width box labeled "REFLECTION COEFFICIENT SCALE"

- Number line from 0.1 to 0.3 (within box bounds)

- Four markers with values and state labels:

- 0.16 / CRIT

- 0.19 / LOW

- 0.20 / NORM

- 0.23 / SURP

Fourth row — Behavioral Effects (506):

- Full-width box labeled "BEHAVIORAL EFFECTS"

- Four effect descriptions:

- 0.16: "Reduced - conserve cognitive resources"

- 0.19: "Slightly reduced - cautious operation"

- 0.20: "Baseline - normal self-observation"

- 0.23: "Enhanced - energy allows exploration"

Caption:

- "Energy-aware regulation ensures efficient cognitive resource management."

- "Low energy reduces self-observation; surplus energy enables exploration."

FIG. 6 — End-to-End Signal Flow with Self-Observation Integration

This figure shows the complete signal flow of the self-sustaining cognitive loop with the self-observation pathway highlighted as architecturally distinct.

Reference numerals: 600, 602, 604, 606, 608, 610

Vertical flow layout:

Top (600):

- Full-width box labeled "ENVIRONMENTAL INPUT"

- Inside: "(Light / Temperature / Pressure)"

- Arrow down to Neuromorphic Core

Second row (602):

- Full-width box labeled "NEUROMORPHIC CORE (SNN)" (bold border)

- Inside: "V = V × leak + I_ext + I_refl + I_syn"

- Inside: "output = mean(V_1, V_2, ..., V_N)"

- Two arrows down to Self-Observation and Cognitive Modulation

Third row (side by side):

- Left (604): Box labeled "SELF-OBSERVATION"

- Inside: "(Patent B - This Patent)"

- Inside: "z^(-1) delay + refl_coeff"

- Right (606): Box labeled "COGNITIVE MODULATION"

- Inside: "refl_coeff adjustment"

- Inside: "Attention / Intention"

- Bidirectional arrows between 604 and 606

Fourth row (right side):

- (608): Box labeled "MOTOR ACTUATION"

- Inside: "Servo Control / Response"

- Arrow down from Cognitive Modulation

Fifth row (right side):

- (610): Box labeled "ENERGY HARVESTING"

- Inside: "Piezo / Thermal Recovery"

- Arrow down from Motor Actuation

Two feedback loops (visually distinguished):

Loop 1 — Self-Observation (Patent B — THIS PATENT):

- Short dashes (bold, stroke-width 3) from Self-Observation (604) back to SNN (602)

- Left side of diagram

- Label: "SELF-OBS LOOP" (rotated vertical text)

Loop 2 — Energy (Patent A):

- Long dashes (stroke-width 2) from Energy Harvesting (610) back to SNN (602)

- Right side of diagram

- Label: "ENERGY LOOP" (rotated vertical text)

Legend box:

- Short dashes (bold) line sample: "Short dashes (bold): Self-Observation (this patent)"

- Long dashes line sample: "Long dashes: Energy feedback (Patent A)"

Caption:

- "The self-observation pathway feeds the network's own aggregate state"

- "back as explicit input at configurable gain (0.0 to 1.0)."

Sheet numbering: "6/6" at top center

GENERAL DRAWING REQUIREMENTS (per 37 CFR 1.84)

- Paper size: 8.5 x 11 inches (US Letter)

- Top margin: 1 inch

- Left margin: 1 inch

- Right margin: 5/8 inch

- Bottom margin: 3/8 inch

- Black lines on white background ONLY

- No color, no grayscale shading (use hatching patterns instead)

- Line weight: sufficiently heavy for adequate reproduction

- Each figure labeled: "FIG. 1", "FIG. 2", etc. - Sheet numbers at top center: "1/6", "2/6", etc.

- No frames or borders around drawing area

- All text in figures must be at least 1/8 inch (3.2 mm) high

- Reference numerals must be used consistently across all figures